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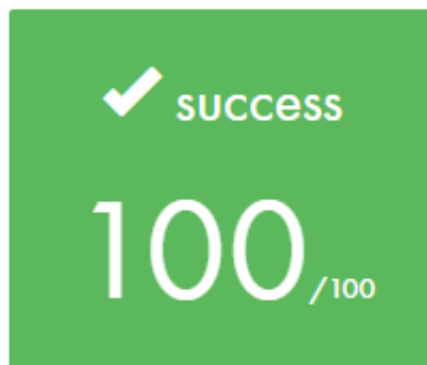
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# Net\_Practice

## 42cursus' project #10

The **net\_practice** project is our first networking related project in 42 curriculum. It consists in 10 exercises in which we should configure different small-scale networks into communicating with each other - all of them using the concepts learnt about TCP/IP addressing.

## Final score



Completed



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## First things first...

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### What is a Network?

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In simple terms, a **Network** in computing is a group of two or more devices that can communicate amongst each other. By communicating we mean the possibility of **sending** and **receiving** data (or *packets*) - between different devices (or *nodes*).

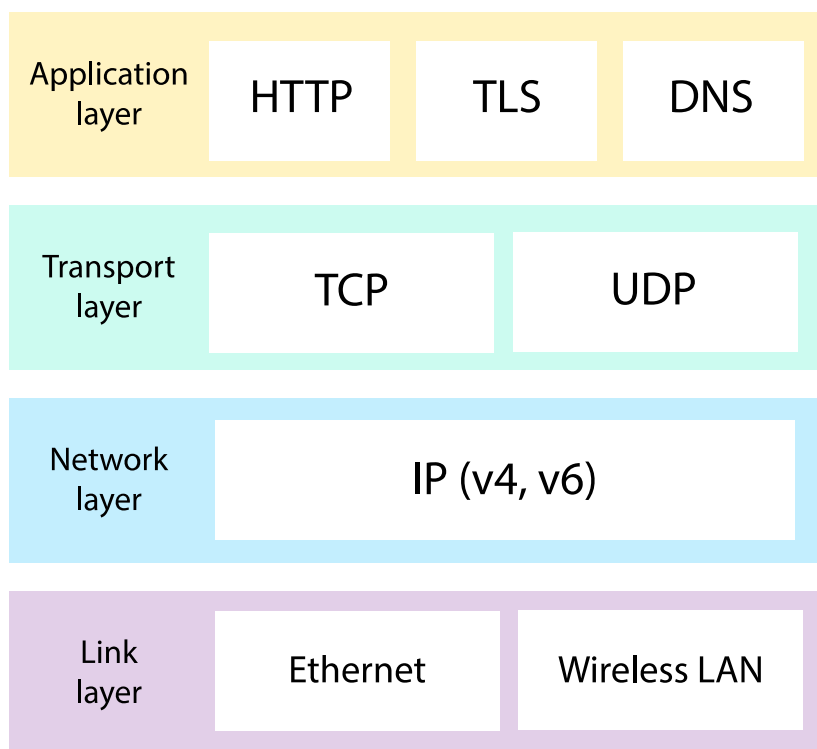
The **Internet** is one of many examples of a network. It is probably the biggest network in the world, and connects many different nodes, making the sharing of data available between devices located anywhere in the world.

The Internet is considered a **Public Network**. A Public Network allows for data transfer between any device connected to it, with no or very little control access. Usually, it is operated by a telecommunications company for the specific purpose of providing data transmission services to the public.

On the other hand, we also have **Private Networks**. A Private Network allows for communication between devices that are restricted to a specific location or domain. The data transfer is not allowed for anyone unregistered, making it a much safer and controlled environment. A Home Network is an example of private network. In there, you can connect to your personal printer when no one who isn't connected to your house's Wi-Fi, for example, could.

## 🔗 What is TCP and IP Address?

In order to transfer data from one point to another across a network, many different types of protocols exist to ensure access, transport, security and stability. All these protocols work together, each one responsible for what we call a "layer" in the communication process.



**Transmission Control Protocol (or TCP)** is a communication standard protocol that enables nodes to communicate with each other. TCP is responsible for "breaking" data into small pieces (or packets), sending them over the network and then "rebuilding" them all back again while ensuring there is no data lost in the process.

And how does it know where to send those pieces to? It does so utilizing an **Internet Protocol Address (or IP Address)**, which is a series of numbers used to identify any device connected to a network, either public or private.

TCP and IP are not the same thing, but rather **two separated protocols** that work together in order ensure data transfer between different devices.

There are two different version of IP Addresses: **IPv4** and **IPv6**. For the purpose of this project, we will only need to know about IPv4, the oldest and more broadly used protocol model. From now on, in this guide, we will use the terms IP and IPv4 interchangeably.

## 🔗 IPv4 and Subnet Masks

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An IPv4 address is a 32-bit number divided into four 8-bit blocks.

Each of theses blocks range from 0 to 255 or, in binary, 00000000 to 11111111 .

It is fundamental to learn how to visualize each IP Address in its binary form. The reason for it is the fact that every IP Address can be split into two separate pieces of information: the **Network** and the **Host** address.

The portion corresponded to the **Network Address** is the identifier of the network in which the devices are connected. In order to communicate with each other, the nodes must be all in the same network, therefore, have the same Network Address portion of their respective IPs.

The portion corresponded to the **Host Address** is the individual identifier of the device. Each node in the same network must have a singular, unique Host Address.

In order to identify which part of the full IP Address correspond to the Network and which correspond to the Host, we musk apply to it a **Network Mask (or Subnet Mask)**.

A Mask is also a 32-bit number divided into four 8-bit blocks. However, its purpose is to identify which bits are actively part of the Network identification. To do so, it marks with 1s the network bits.

There are two ways to represent a mask: by a full IP Mask or by CIDR.

Let's look at an example:

```
IP Address:    153.172.250.12
Subnet Mask:   255.255.255.0
CIDR:         /24
```



In binary, respectively, we have:

```
IP Address:    10011001.10101100.11111010.00001100
Subnet Mask:   11111111.11111111.11111111.00000000
```



According to the mask, the first 24-bits are marked as part of the Network Address. That means that the first 24-bits of the IP Address represent the Network Address.

```
Network Address: 10011001.10101100.11111010.????????
or:              153.172.250.?
IP/CIDR:         153.172.250.12/24
```



In practical terms, we can say that any device with IPs ranging from 192.172.250.0 to 192.172.250.255 are in the same network and can, therefore, communicate with each other freely.

Let's look at another example:

```
IP Address:    153.172.175.12
Subnet Mask:   255.255.240.0
CIDR:         /20
```



In binary, respectively, we have:

```
IP Address:    10011001.10101100.10101111.00001100
Subnet Mask:   11111111.11111111.11110000.00000000
```



According to the mask, now the first 20-bits are marked as part of the Network Address. That means that the first 20-bits of the IP Address represent the Network Address.

```
Network Address: 10011001.10101100.1010?????.?????????  
or:             153.172.(160-175).(0-255)  
IP/CIDR:        153.172.175.12/20
```



In practical terms, we can say that any device with IPs ranging from 192.172.160.0 to 192.172.175.255 are in the same network and can, therefore, communicate with each other freely.

It's important to notice that in the network mask, when reading it from left to right, once a zero appears, the mask will be completed with only zeros. That means 255.255.240.0 is a valid mask, but 255.255.240.128 isn't. For masks, there are only nine possible 8-bit blocks:

```
0 0 0 0 0 0 0 0 -> 0  
1 0 0 0 0 0 0 0 -> 128  
1 1 0 0 0 0 0 0 -> 192  
1 1 1 0 0 0 0 0 -> 224  
1 1 1 1 0 0 0 0 -> 240  
1 1 1 1 1 0 0 0 -> 248  
1 1 1 1 1 1 0 0 -> 252  
1 1 1 1 1 1 1 0 -> 254  
1 1 1 1 1 1 1 1 -> 255
```



However, something else that is important to understand is that, in this process of dividing a larger network into smaller ones or, subnets, we must reserve 2 IP addresses that cannot be used by any device.

The **first IP in the range** is reserved to identify the subnet.

The **last IP in the range** is reserved for broadcasting messages across all devices in the subnet.

In the first example, the IPs 192.172.250.0 and 192.172.250.255 would be reserved. So, only the IPs from 192.172.250.1 to 192.172.250.254 could be assigned for other devices.

That means, in practical terms, that we must choose network masks carefully, having in mind how many devices you plan to connect to your network.

| <b>CIDR</b> | <b>SUBNET MASK</b> | <b>WILDCARD MASK</b> | <b># OF IP ADDRESSES</b> | <b># OF ADDRESSES</b> |
|-------------|--------------------|----------------------|--------------------------|-----------------------|
| /32         | 255.255.255.255    | 0.0.0.0              | 1                        | 1                     |
| /31         | 255.255.255.254    | 0.0.0.1              | 2                        | 2                     |
| /30         | 255.255.255.252    | 0.0.0.3              | 4                        | 2                     |
| /29         | 255.255.255.248    | 0.0.0.7              | 8                        | 6                     |
| /28         | 255.255.255.240    | 0.0.0.15             | 16                       | 14                    |
| /27         | 255.255.255.224    | 0.0.0.31             | 32                       | 30                    |
| /26         | 255.255.255.192    | 0.0.0.63             | 64                       | 62                    |
| /25         | 255.255.255.128    | 0.0.0.127            | 128                      | 126                   |
| /24         | 255.255.255.0      | 0.0.0.255            | 256                      | 254                   |
| /23         | 255.255.254.0      | 0.0.1.255            | 512                      | 510                   |
| /22         | 255.255.252.0      | 0.0.3.255            | 1,024                    | 1,022                 |
| /21         | 255.255.248.0      | 0.0.7.255            | 2,048                    | 2,046                 |
| /20         | 255.255.240.0      | 0.0.15.255           | 4,096                    | 4,094                 |
| /19         | 255.255.224.0      | 0.0.31.255           | 8,192                    | 8,190                 |
| /18         | 255.255.192.0      | 0.0.63.255           | 16,384                   | 16,382                |
| /17         | 255.255.128.0      | 0.0.127.255          | 32,768                   | 32,766                |
| /16         | 255.255.0.0        | 0.0.255.255          | 65,536                   | 65,534                |
| /15         | 255.254.0.0        | 0.1.255.255          | 131,072                  | 131,070               |
| /14         | 255.252.0.0        | 0.3.255.255          | 262,144                  | 262,142               |
| /13         | 255.248.0.0        | 0.7.255.255          | 524,288                  | 524,286               |

|     |             |                 |               |      |
|-----|-------------|-----------------|---------------|------|
| /12 | 255.240.0.0 | 0.15.255.255    | 1,048,576     | 1,04 |
| /11 | 255.224.0.0 | 0.31.255.255    | 2,097,152     | 2,09 |
| /10 | 255.192.0.0 | 0.63.255.255    | 4,194,304     | 4,19 |
| /9  | 255.128.0.0 | 0.127.255.255   | 8,388,608     | 8,38 |
| /8  | 255.0.0.0   | 0.255.255.255   | 16,777,216    | 16,7 |
| /7  | 254.0.0.0   | 1.255.255.255   | 33,554,432    | 33,5 |
| /6  | 252.0.0.0   | 3.255.255.255   | 67,108,864    | 67,1 |
| /5  | 248.0.0.0   | 7.255.255.255   | 134,217,728   | 134, |
| /4  | 240.0.0.0   | 15.255.255.255  | 268,435,456   | 268, |
| /3  | 224.0.0.0   | 31.255.255.255  | 536,870,912   | 536, |
| /2  | 192.0.0.0   | 63.255.255.255  | 1,073,741,824 | 1,07 |
| /1  | 128.0.0.0   | 127.255.255.255 | 2,147,483,648 | 2,14 |
| /0  | 0.0.0.0     | 255.255.255.255 | 4,294,967,296 | 4,29 |

## 🔗 Connecting multiple devices

As it was previously said, Public Networks like the **Internet** are established by some telecommunication provider in order to connect devices from all around the globe.

A **Public IP Address** is assigned by your **Internet Service Provider (or ISP)**, directly into your personal **router**, and allows connectivity directly to your personal, private network.

Your private network is established by a **switch** located inside the router provided to you by your ISP. Your router is responsible for assigning a **Private IP Address** directly to any device connected to your private network, following the previously mentioned rules regarding subnet masks and IPs.



There are some IP addresses that are reserved for specific uses. Therefore, when assigning a IP Address inside of a network connected to the internet, you must pay attention to the list below:

| IP Address Range               | Reserved                                     |
|--------------------------------|----------------------------------------------|
| 10.0.0.0 - 10.255.255.255      | reserved for private IPs (Class A)           |
| 127.0.0.0 -<br>127.255.255.255 | reserved for loopback and internal testing   |
| 172.16.0.0 - 172.31.255        | reserved for private IPs (Class B)           |
| 192.168.0.0 - 192.168.0.0      | reserved for private IPs (Class C)           |
| 224.0.0.0 –<br>239.255.255.255 | reserved for multicast                       |
| 240.0.0.0 –<br>255.255.255.255 | reserved for experimental, used for research |

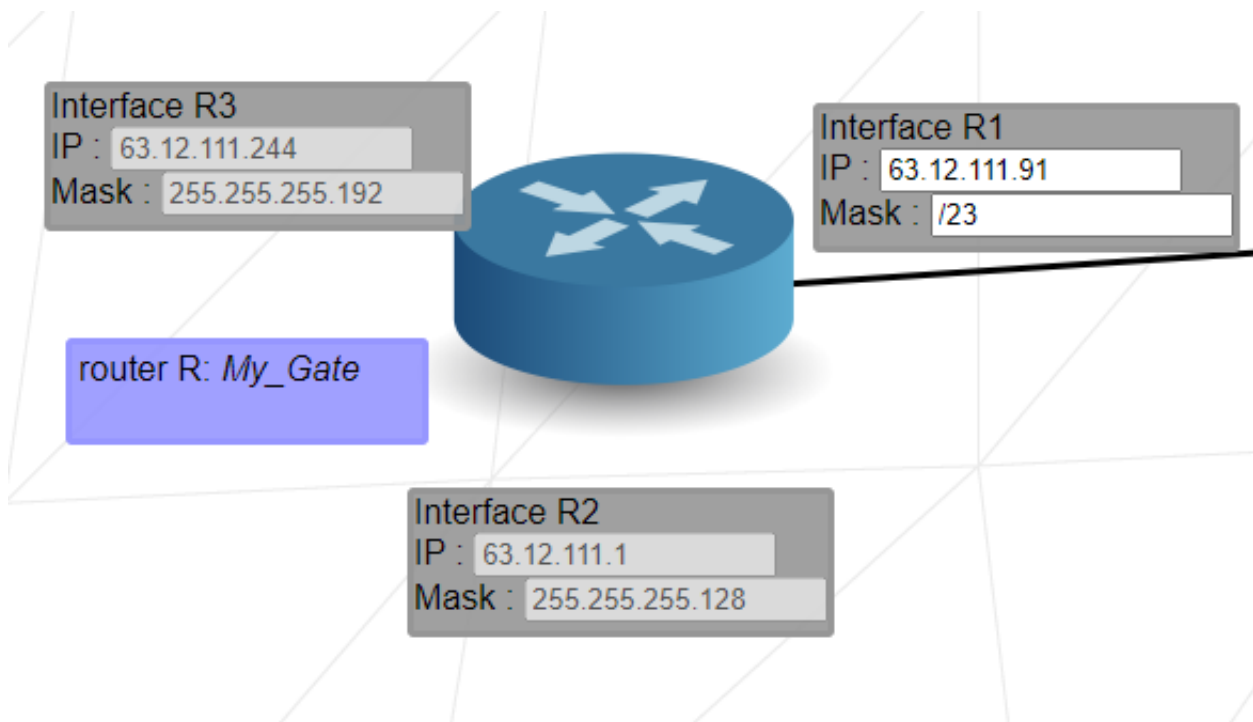
## 🔗 Switch

A network **switch** is responsible for distributing packets between devices within the same network - usually, a local one (called LAN). A switch does not have any interface and it cannot talk to a network outside of its own.



## 🔗 Router

A **router** is responsible for connecting multiple networks together. Every router has an interface for every network it connects it to.



Since it connects multiple networks together, the range of possible IP addresses on one of its interfaces must never overlap the range of its other interfaces. An overlap of ranges would imply the interfaces belong to the same network.

## 🔗 Routing Table

A **Routing Table** is a simple data table stored in a router or network host that lists all the routes to a particular network destination.

| Routes :        |                   |
|-----------------|-------------------|
| 49.198.111.0/22 | => 163.172.250.12 |
| 10.0.0.0/27     | => 163.172.250.12 |
| default         | => 163.172.250.12 |

In Net Practice, a routing table consists of only two simple informations:

- **Destination** (on the left): it's the IP address that you want to send a package to, combined with the CIDR of that network: 190.3.2.252/30. If you don't want to specify a destination, or you want to make it available for the entire network, you can just set it to default or 0.0.0.0/0.
- **Next Hop** (on the right): it's the IP address of the next router that you need to send the packages to in order to reach the destination-network.

# Net Practice

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▶ Level 1

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▶ Level 2

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▶ Level 3

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▶ Level 4

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▶ Level 5

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▶ Level 6

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▶ Level 7

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▶ Level 8

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▶ Level 9

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▶ Level 10